

Time allowed: 85 minutes
All questions are of equal value.
Attempt all questions.
Calculators may be used.

QUESTION 1 (START A NEW PAGE)

(a) Integrate with respect to x :

(i) $\frac{x^2+3}{2x}$

(ii) $\cos \frac{1}{3}x$

(iii) $(\sqrt{x} + 4)^2$

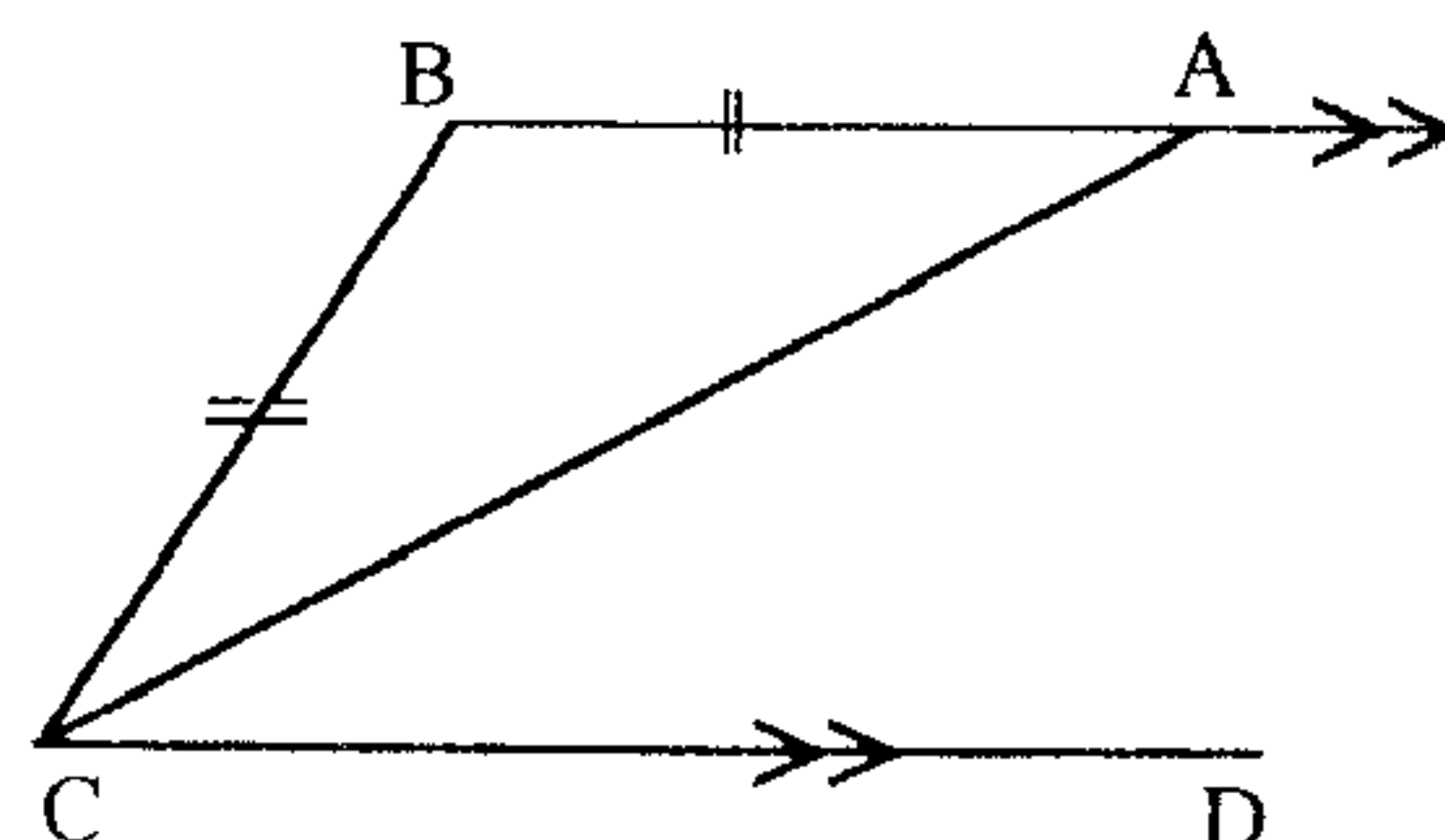
(b) Evaluate: (i) $\int_1^3 \frac{1}{2x-1} dx$.

(ii) $\int_{\frac{\pi}{8}}^{\frac{\pi}{6}} \sec^2 2x dx$.

(c) Find the general term of the arithmetic sequence $\{101, 97, 93, \dots\}$.

QUESTION 2 (START A NEW PAGE)

(a) Given $BA \parallel CD$ and $BA = BC$, prove that AC bisects $\angle BCD$.



(b) Find the volume of the solid formed when the area bounded by $y = \sqrt{4-x^2}$ and the x -axis for $-1 \leq x \leq 2$ is rotated one revolution about the x -axis.

(c) Sketch the following curves showing their intercepts with the co-ordinate axes and any asymptotes if they exist.

(i) $y = \frac{x+2}{x-1}$.

(ii) $x^2 + y^2 - 6x + 8y = 0$.

QUESTION 3 (START A NEW PAGE)

(a) Mrs Jones decides to set up a small fund to pay for her Christmas holiday. Each weekend she keeps \$25 of her ironing money and deposits the \$25 in her fund at the start of that week. The fund receives 3.25% p.a. interest and the interest is compounded at the end of each week. (Assume 1 year = 52 weeks)

(i) Find the value of her first deposit at the end of 50 weeks. (Give answer to nearest cent)

(ii) Find the value of her fund at the end of 50 weeks. (Give answer to nearest dollar)

(b) (i) Show that the equation of the tangent to $y = e^{3x}$ at the point where $x = 1$ is given by $y = e^3(3x - 2)$.

(ii) Draw a diagram showing the area bounded by $y = e^{3x}$, the tangent in (i) and the y -axis.

(iii) Find the exact area of the region described in (ii).

QUESTION 4 (START A NEW PAGE)

(a) If $f'(x) = \cot x + x$ and $f\left(\frac{\pi}{2}\right) = 0$, find an expression for $f(x)$.

(b) Find the area bounded by $y = \sqrt{3-x}$ and the co-ordinate axes.

(c) (i) Draw a neat sketch showing the area bounded by the curve $y = 1 + \sqrt{x}$ and the x -axis for $0 \leq x \leq 4$.

(ii) Find the volume of the solid formed when the area in (i) is rotated one revolution about the y -axis.

QUESTION 5

(START A NEW PAGE)

- (a) ABCD is a parallelogram and P is a point on the diagonal BD.

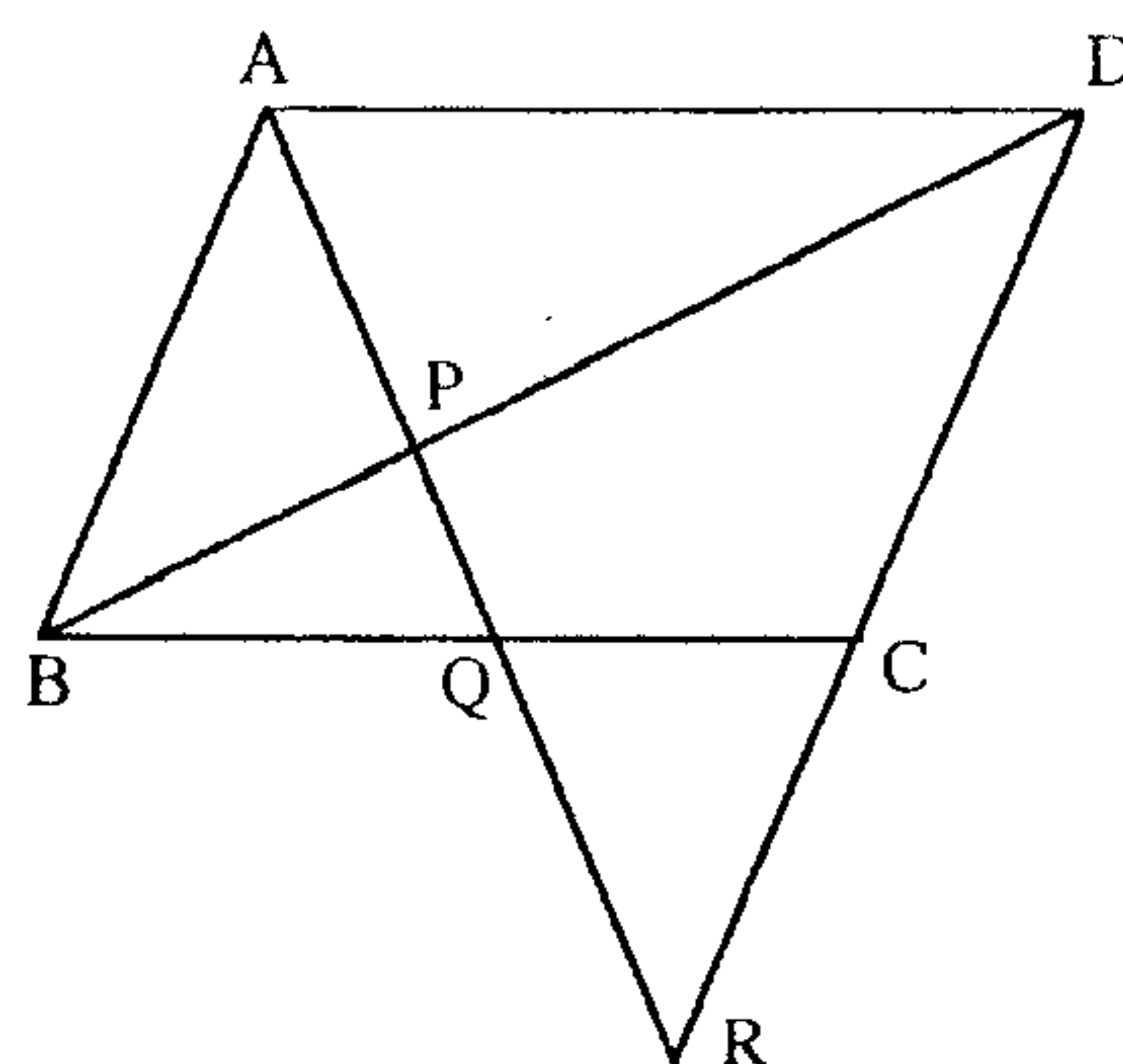
(i) Prove that $\triangle APD$ and $\triangle QPB$ are similar.

If the ratio $AP:PQ = 2:1$

(ii) Prove that Q is the midpoint of BC.

(iii) Prove that $\triangle ABQ$ and $\triangle RCQ$ are congruent.

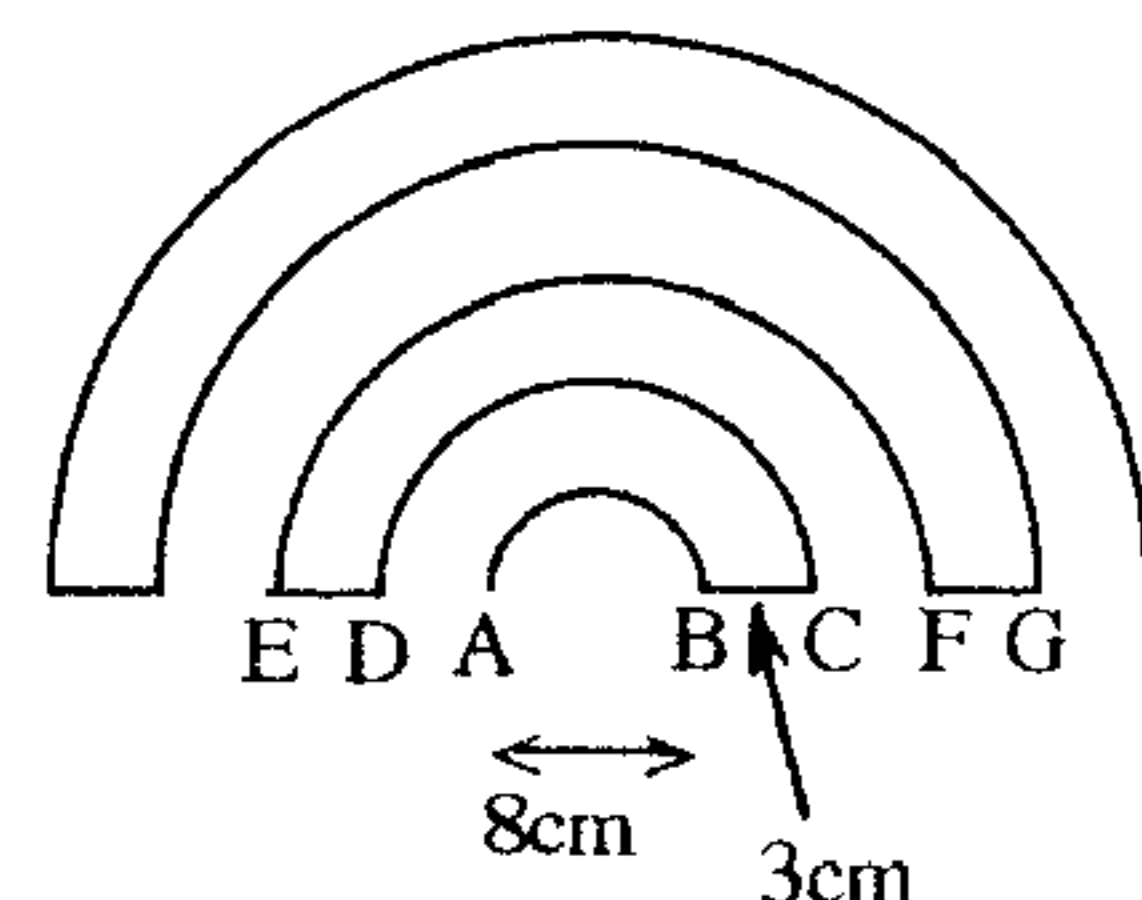
(iv) Prove that C is the midpoint of DR.



- (b) A piece of wire (ABCDEF...) is used to make a pattern formed from many concentric semi-circular arcs (AB, CD, EF, ...) joined by short intervals (BC, DE, FG, ...). The diameter of the smallest semi-circle is 8cm and each interval is 3cm. (see diagram)

(i) What is the length of the 20th semi-circular arc?

(ii) How much wire is needed to form a pattern containing 20 semi-circular arcs?



THIS IS THE END OF THE PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

QUESTION 1

(i) $\int \frac{1}{2}x + \frac{1}{2x} dx = \frac{1}{4}x^2 + \frac{1}{2}\ln x + c$

(ii) $\int \cos \frac{1}{3}x dx = 3\sin \frac{1}{3}x + c$

(iii) $\int x + 8\sqrt{x} + 16 dx = \frac{1}{2}x^2 + \frac{16}{3}x^{3/2} + 16x + c$

(i) $\int_1^3 \frac{1}{2x-1} dx = \left[\frac{1}{2} \ln(2x-1) \right]_1^3$
 $= \frac{1}{2} \ln 5 - \frac{1}{2} \ln 1$
 $= \frac{1}{2} \ln 5$

(ii) $\int_{\pi/8}^{\pi/6} \sec^2 2x dx = \left[\frac{1}{2} \tan 2x \right]_{\pi/8}^{\pi/6}$
 $= \frac{1}{2} \tan \pi/3 - \frac{1}{2} \tan \pi/4$
 $= \frac{\sqrt{3}}{2} - \frac{1}{2}$

$a = 101$ $d = -4$

$T_n = 101 + (n-1)(-4)$

$= 105 - 4n$

QUESTION 2

Let $\hat{BAC} = x^\circ$

$\hat{BCA} = x^\circ$ (equal angles are opposite equal sides)

$\hat{ACD} = x^\circ$ (ASHCD, alternate angles)

$\hat{ACB} = \hat{ACD}$ (both x°)

$\therefore AC$ bisects $\hat{BCD}$ ($\hat{ACB} = \hat{ACD}$)

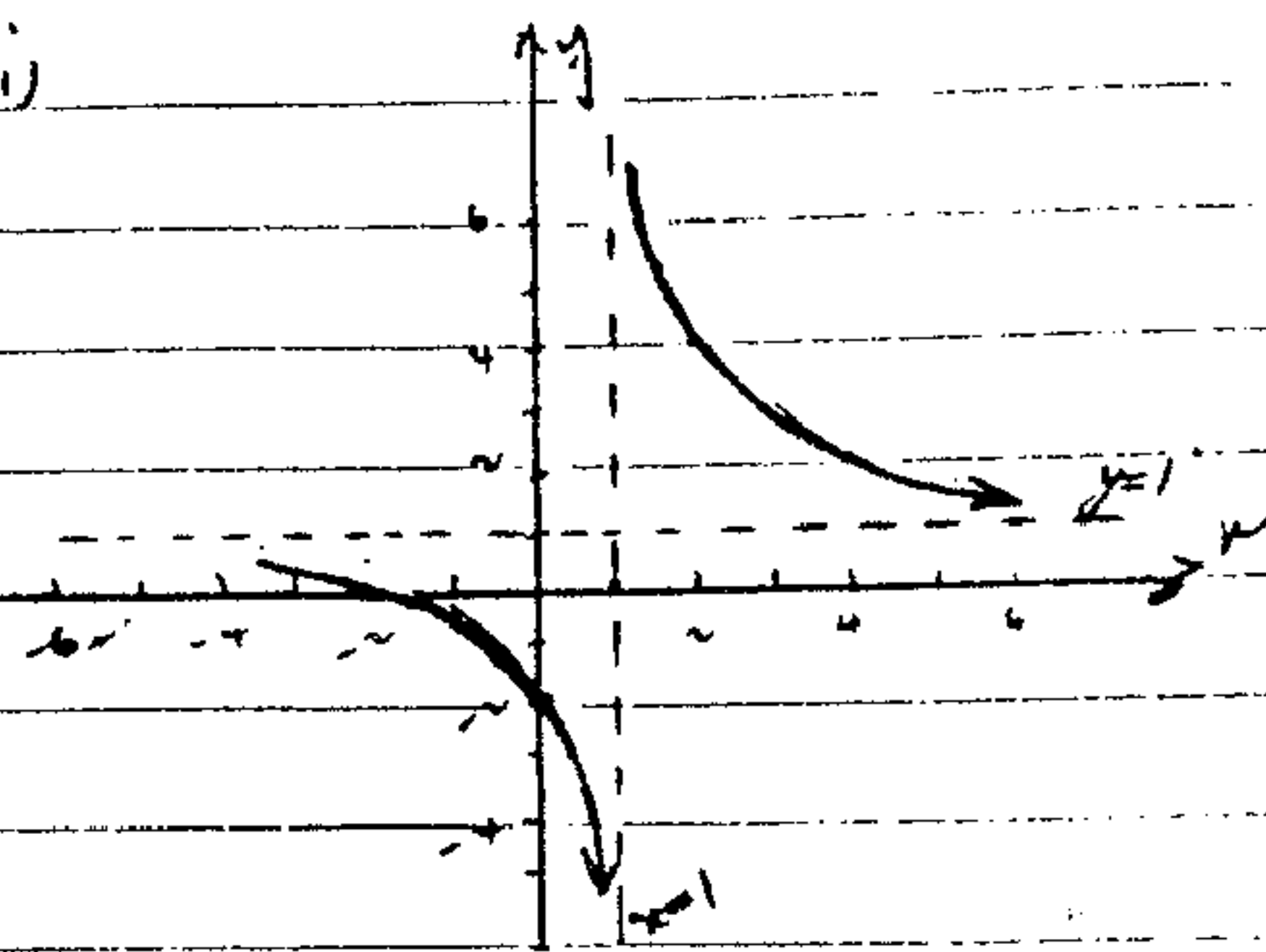
$V = \pi \int_{-1}^2 y^2 dx$

$= \pi \int_{-1}^2 (4-x^2) dx$

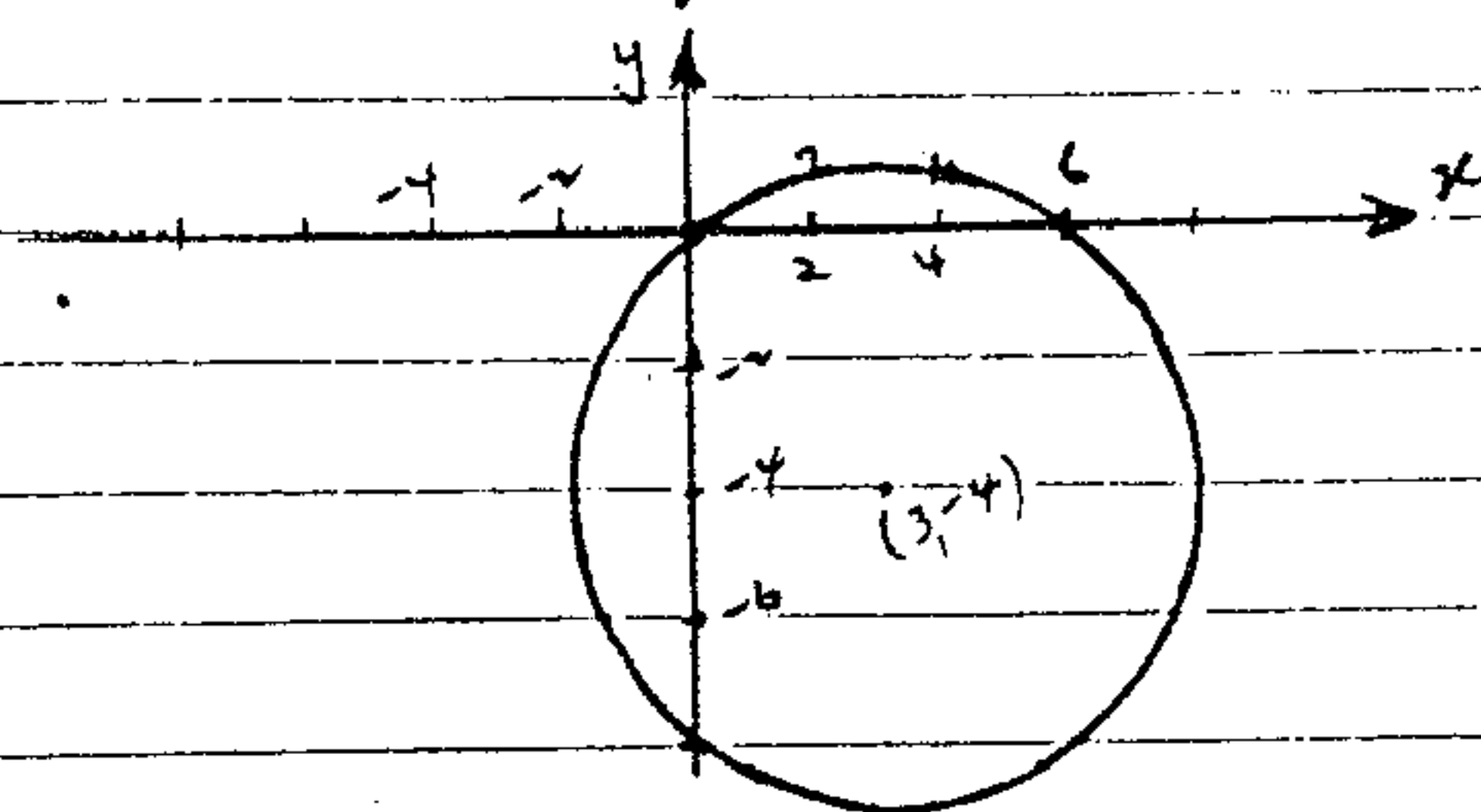
$= \pi \left[4x - \frac{1}{3}x^3 \right]_{-1}^2$

$= 9\pi$ cubic units

(c)(i)



(ii) $(x-3)^2 + (y+4)^2 = 25$



QUESTION 3

(a)(i) Value = $\$ 25 \times \left(1 + \frac{3.25}{5200} \right)^{50}$
 $= \$ 25.79$

(ii) Value = $\$ 25 \left(1.000625^{50} + 1.000625^{49} + \dots + 1.000625 \right)$
 $= \$ 25 \times 1.000625 \frac{(1.000625^{50} - 1)}{1.000625 - 1}$
 $= \$ 1270$

(b)(i) $y' = 3e^{3x}$

at $x=1$, $y' = 3e^3$

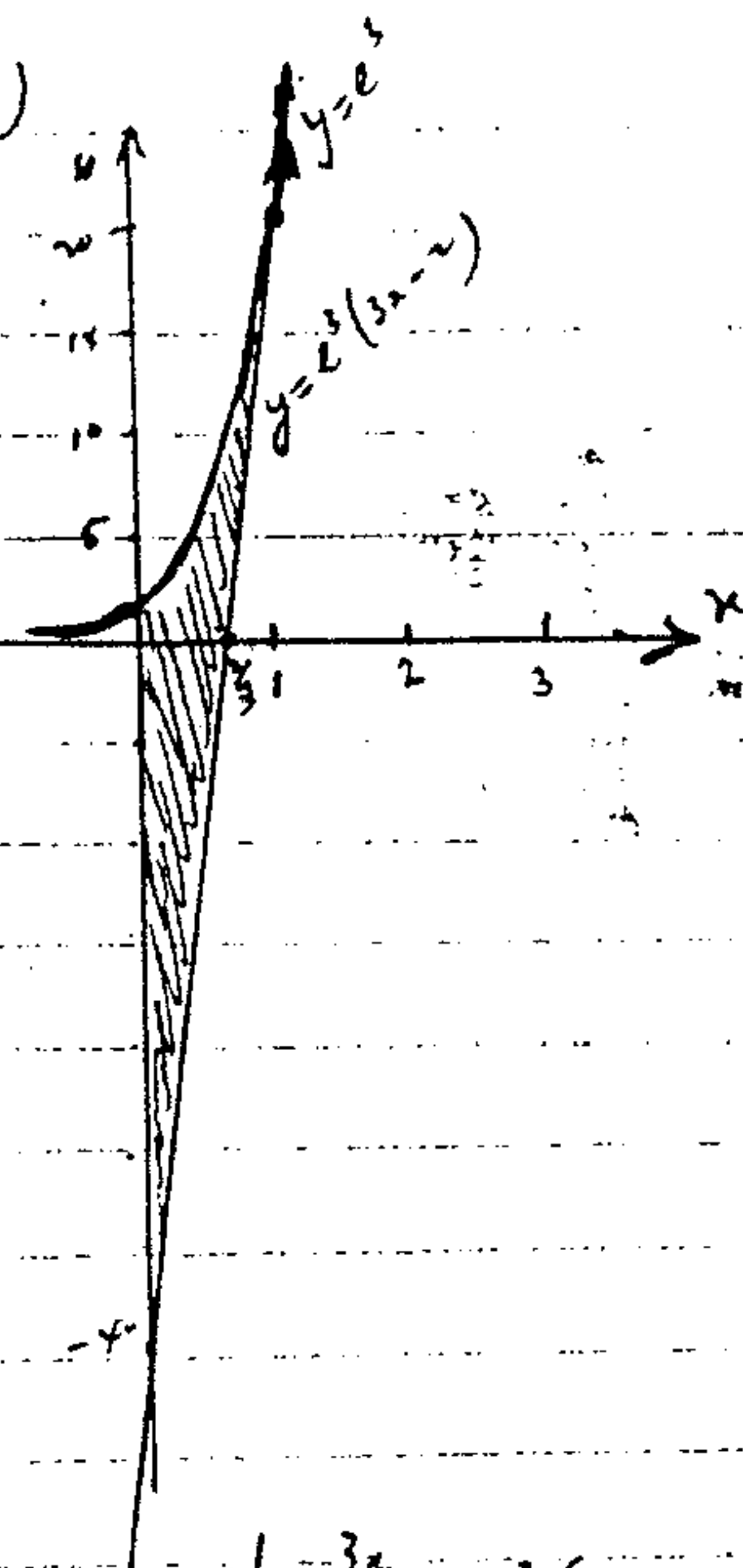
$y = e^3$

tangent: $y - e^3 = 3e^3(x-1)$

$y = 3e^3x - 2e^3$

$y = e^3(3x-2)$

Q3(cont)
3(b)(ii)



(iii) $A = \int_0^1 e^{-3x} - e^3(3x-2) dx$
 $= \left[-\frac{1}{3}e^{-3x} - \frac{3}{2}e^3x^2 + 2e^3x \right]_0^1$

$A = \frac{1}{6}(5e^3 - 2)$ square units

QUESTION 4

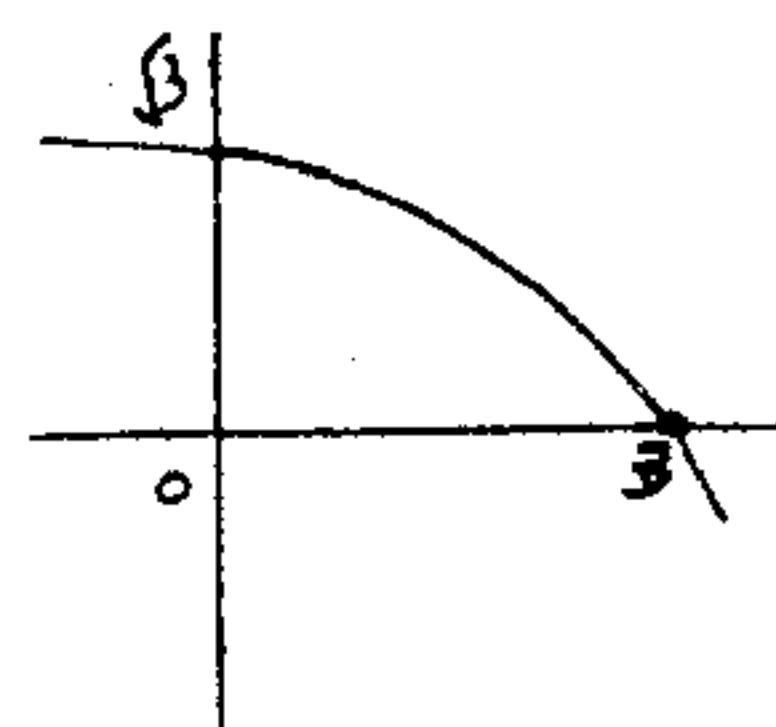
(a) $f'(x) = \frac{\cos x}{\sin x} + x$

$f(x) = \ln(\sin x) + \frac{1}{2}x^2 + c$

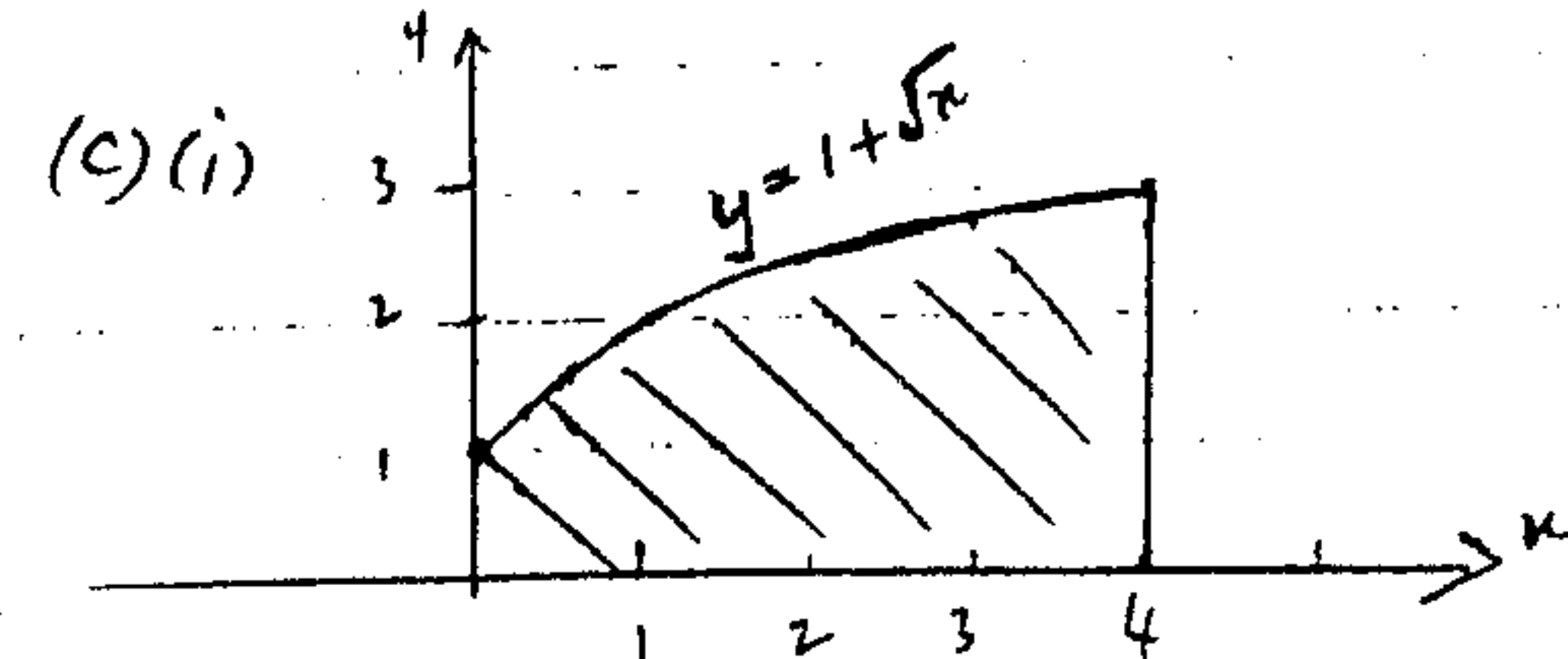
$f(\pi/2) = 0 \Rightarrow 0 = \ln(\sin \pi/2) + \frac{\pi^2}{8} + c$
 $c = -\frac{\pi^2}{8}$

$\therefore f(x) = \ln(\sin x) + \frac{1}{2}x^2 - \frac{\pi^2}{8}$

(b) $A = \int_0^3 (3-x)^{1/2} dx$
 $= \left[\frac{2}{3}(3-x)^{3/2} \right]_0^3$



$= 2\sqrt{3}$ square units.



(ii) $V = \pi \int_0^3 16 dy - \pi \int_0^3 (y-1)^4 dy$
 $= \pi \left[16y \right]_0^3 - \pi \left[\frac{1}{5}(y-1)^5 \right]_0^3$
 $= 48\pi - \frac{32}{5}\pi$
 $= \frac{208}{5}\pi$ cubic units

QUESTION 5

(a)(i) $\angle APD + \angle QPB$

$\hat{APD} = \hat{QPB}$ (vertically opposite angles)

$\hat{ADP} = \hat{QBP}$ ($AD \parallel BC$, opposite sides of para, alternate angles)

$\therefore \triangle APD \parallel \triangle QPB$ (equiangular)

(ii) $\frac{BQ}{DA} = \frac{PQ}{AP}$ (ratio of corresponding sides)
 $= \frac{1}{2}$

$\therefore BQ = \frac{1}{2}AD$

but $BC = AD$ (opposite sides of para)

$\therefore BQ = \frac{1}{2}BC$

$\therefore Q$ is midpt of BC

(iii) $\angle AQR$ and $\angle RQR$

$BQ = QC$ (proven in (ii))

5 (cont)

(iii) $\hat{AOB} = \hat{COE}$ (vertically opposite angles)

$\hat{BAO} = \hat{QCE}$ (AB/QR opposite sides of para, alternate angles are equal)

$\therefore \triangle ABO \cong \triangle QCE$ (AAS)

(iv) $AB = CR$ (corresponding sides in congruent triangles)

$AB = DC$ (opposite sides of para.)

$\therefore DC = CR$ (both = AB)

$\therefore C$ is midpoint of DR.

5) (i) $\{4\pi, 7\pi, 10\pi, \dots\}$ $a = 4\pi, d = 3\pi$

$$l = 4\pi + 19 \times 3\pi \quad n = 19$$

$$\text{length} = 61\pi \text{ cm}$$

$$(ii) l = \frac{20}{2} \{8\pi + 19(3\pi)\} + 19 \times 3$$

$$= 10(65\pi) + 57 \text{ cm}$$

$$\text{length} = 650\pi + 57 \text{ cm}$$